

Ceretam®

| Ceretam® 800mg Film-Coated Tablet | Ceretam® Oral Solution 20%w/v

Description
Ceretam® 800mg Film-Coated Tablet: Each Ceretam tablet contains piracetam 800 mg. It is a white to off-white, oblong shaped, film-coated tablet with CC logo and embossed line on both sides of the tablet.
Ceretam® Oral Solution 20%w/v: Each 5ml contains piracetam 1g. It is a clear solution with strawberry and banana flavours. Preservatives methylparaben 0.135%w/v and propylparaben 0.015%w/v.

Pharmacodynamics
Pharmacotherapeutic group: Nootropics, ATC code: N06B X03
Mechanism of action: Piracetam's mode of action in cortical myoclonus is as yet unknown.

Pharmacodynamic effects: Piracetam exerts its haemorrhological effects on the platelets, red blood cells, and vessel walls by increasing erythrocyte deformability and by decreasing platelet aggregation, erythrocyte adhesion to vessel walls and capillary vasospasm.

- Effects on the red blood cells: In patients with sickle cell anaemia, piracetam improves the deformability of the erythrocyte membrane, decreases blood viscosity, and prevents rouleaux formation.
- Effects on platelets: In healthy volunteers and in patients with Raynaud's phenomenon, increasing doses of piracetam up to 12 g was associated with a dose-dependent reduction in platelet functions compared with pre-treatment values (tests of aggregation induced by ADP, collagen, epinephrine and BTG release), without significant change in platelet count. Piracetam prolonged bleeding time.
- Effects on blood vessels: In animal studies, piracetam inhibited vasospasm and counteracted the effects of various spasmogenic agents. It lacked any vasodilatory action and did not induce "steal" phenomenon, nor low or no reflow, nor hypotensive effects.

In healthy volunteers, piracetam reduced the adhesion of RBCs to vascular endothelium and possessed also a direct stimulant effect on prostacycline synthesis in healthy endothelium.

- Effects on coagulation factors: In healthy volunteers, compared with pre-treatment values, piracetam up to 9.6 g reduced plasma levels of fibrinogen and von Willebrand's factors (VIII : C; VIII R : AG; VIII R : vW) by 30 to 40 %, and increased bleeding time.

In patients with both primary and secondary Raynaud phenomenon, compared with pre-treatment values, piracetam 8 g/d during 6 months reduced plasma levels of fibrinogen and von Willebrand's factors (VIII : C; VIII R : AG; VIII R : vW (RCF)) by 30 to 40 %, reduced plasma viscosity, and increased bleeding time.

Pharmacokinetics

Absorption
Piracetam is rapidly and almost completely absorbed. Peak plasma levels are reached within 1.5 hours after administration. The extent of oral bioavailability, assessed from the Area Under Curve (AUC), is close to 100% for capsules, tablets and solution. Peak levels and AUC are proportional to the dose given.

Distribution
The volume of distribution of piracetam is 0.7 L/kg. Piracetam crosses the blood-brain and the placental barriers and diffuses across membranes used in renal dialysis.

Biotransformation
Up to now, no metabolite of piracetam has been found.

Elimination
Piracetam is excreted almost completely in urine and the fraction of the dose

excreted in urine is independent of the dose given. Excretion half-life values are consistent with those calculated from plasma / blood data. The plasma half-life is 5.0 hours, in young adult men. Clearance of the compound is dependent on the renal creatinine clearance and would be expected to diminish with renal insufficiency.

Indications

1. Treatment of the elderly with some degree of cerebral functional impairment such as loss of memory, a lack of concentration or alertness and vertigo.
2. Ceretam is indicated for patients suffering from myoclonus of cortical origin, irrespective of aetiology and should be used in combination with other anti-myoclonic therapies.

Route of Administration

Oral formulations. For oral use.

Dosage and Administration

Oral formulations

Piracetam may be taken with or without food. The film-coated tablets should be swallowed with liquid. It is recommended to take the daily dose in two to four sub-doses.

Parenteral formulations

When parenteral administration is needed (e.g. swallowing difficulties, unconsciousness) piracetam can be administered intravenously at the same recommended daily dose.

- The solution for injection will be administered intravenously over several minutes.
- The solution for infusion will be administered continuously at the recommended daily dose over a 24 hour period.

Adults

The daily dosage should begin at 7.2g, increasing by 4.8g every three or four days up to a maximum of 24g, in two to four divided doses. Treatment with other anti-myoclonic medicinal products should be maintained at the same dosage. Depending on the clinical benefit obtained, the dosage of other such medicinal products should be reduced, if possible.
Once started, treatment with piracetam should be continued for as long as the original cerebral disease persists. In patients with an acute episode, spontaneous evolution may occur over time and an attempt should be made every 6 months to decrease or discontinue the medicinal treatment. This should be done by reducing the dose of piracetam by 1.2g every two days (every three or four days in the case of a Lance- Adams syndrome, in order to prevent the possibility of sudden relapse or withdrawal seizures).

Elderly

Adjustment of the dose is recommended in elderly patients with compromised renal function (see Section Warnings and Precautions; Renal impairment below). For long term treatment in the elderly, regular evaluation of the creatinine clearance is required to allow dosage adaptation if needed.

Renal impairment

The daily dose must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (Cl_{cr}) in ml/min is needed. The Cl_{cr} in ml/min may be estimated from serum creatinine (mg/dl) determination using the following formula:

$$Cl_{cr} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)} \times (0.85 \text{ for women})}{72 \times \text{serum creatinine (mg/dl)}}$$

Group	Creatinine Clearance (ml/min)	Posology and frequency
Normal	>80	usual daily dose, 2 to 4 divided doses
Mild	50-79	2/3 usual daily dose, divided in 2 or 3 doses
Moderate	30-49	1/3 usual daily dose, divided in 2 doses
Severe	<30	1/6 usual daily dose, 1 single intake
End-stage renal disease	-	contraindicated

Hepatic impairment:

No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of dose is recommended (see dose adjustment in Renal Impairment above).

Contraindications

Hypersensitivity to the active substance piracetam, or to any of the excipients of the product or other pyrrolidone derivatives.

Piracetam is contra-indicated in patients with severe renal impairment (renal creatinine clearance of less than 20 ml per minute). It is also contraindicated in patients with cerebral haemorrhage and in patients suffering from Huntington's Chorea.

Warnings & Precautions

Effects on platelet aggregation: Due to the effect of piracetam on platelet aggregation, caution is recommended in patients with severe haemorrhage, patients at risk of bleeding such as gastrointestinal ulcer, patients with underlying disorders of haemostasis, patients with history of haemorrhagic cerebro-vascular accident (CVA), patients undergoing major surgery including dental surgery, and patients using anticoagulants or platelet antiaggregant drugs including low dose acetylsalicylic acid.

Renal insufficiency: Piracetam is eliminated via the kidneys and care should thus be taken in cases of renal insufficiency

Elderly: For long-term treatment in the elderly, regular evaluation of the creatinine clearance is required to allow dosage adaptation if needed.

Discontinuation: Abrupt discontinuation of treatment should be avoided as this may induce myoclonic or generalised seizures in some myoclonic patients.

Use in Pregnancy & Lactation

Pregnancy: There is no adequate data from the use of piracetam in pregnant women. Piracetam crosses the placental barrier. Drug levels in the newborn are approximately 70% to 90% of maternal levels. Piracetam should not be used during pregnancy unless clearly necessary, when benefit exceeds the risks and the clinical condition of the pregnant mother requires treatment with piracetam.

Breast-feeding: Piracetam is excreted in human breast milk. Therefore, piracetam should not be used during breastfeeding or breastfeeding should be discontinued, while receiving treatment with piracetam. A decision must be made whether to discontinue breast-feeding or to discontinue piracetam therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Effects on Ability To Drive and Use Machines

In clinical studies, at dosages between 1.6 - 15 grams per day, hyperkinesia, somnolence, nervousness and depression were reported more frequently in patients on piracetam than on placebo. There is no experience on driving ability in dosages between 15 and 20 grams daily. Caution should therefore be exercised by patients intending to drive or use machinery whilst taking piracetam.

Side-effects

Undesirable effects are listed in the following table per System Organ Class and per frequency. The frequency is defined as follows: very common; common; uncommon; rare; very rare.

Blood and lymphatic system disorders: Not known: haemorrhagic disorder

Immune system disorders: Not known: anaphylactoid reaction, hypersensitivity

Psychiatric disorders: Common: nervousness; Uncommon: depression; Not known: agitation, anxiety, confusion, hallucination

Nervous system disorders: Common: hyperkinesia; Uncommon: somnolence

Not known: ataxia, balance impaired, epilepsy aggravated, headache, insomnia,

Ear and labyrinth disorders: Not known: vertigo

Gastrointestinal disorders: Not known: abdominal pain, abdominal pain upper, diarrhoea, nausea, vomiting

Skin and subcutaneous tissue disorders: Not known: angioneurotic oedema, dermatitis, pruritus, urticaria

General disorders and administration site conditions: Uncommon: asthenia

Investigations: Common: weight increased

Interactions with Other Medicaments

Pharmacokinetics interactions: The drug interaction potential resulting in changes of piracetam pharmacokinetics is expected to be low because approximately 90% of the dose of piracetam is excreted in the urine as unchanged drug.

Thyroid hormones: Confusion, irritability and sleep disorder have been reported during concomitant treatment with thyroid extract (T3 + T4).

Acenocoumarol: In patients with severe recurrent venous thrombosis, piracetam 9.6 g/d did not modify the doses of acenocoumarol necessary to reach INR 2.5 to 3.5, but compared with the effects of acenocoumarol alone, the addition of piracetam 9.6 g/d significantly decreased platelet aggregation, β -thromboglobulin release, levels of fibrinogen and von Willebrand's factors (VIII : C; VIII : vW : Ag; VIII : vW : RCo) and whole blood and plasma viscosity.

Antiepileptic drugs: A 20 g daily dose of piracetam over 4 weeks did not modify the peak and trough serum levels of antiepileptic drugs (carbamazepine, phenytoin, phenobarbitone, valproate) in epileptic patients who were receiving stable doses.

Alcohol: Concomitant administration of alcohol had no effect on piracetam serum levels and alcohol levels were not modified by a 1.6 g oral dose of piracetam.

Symptoms and Treatment of Overdose

No additional adverse event specifically related to overdose has been reported with piracetam.

In acute, significant overdosage, the stomach may be emptied by gastric lavage or by induction of emesis. There is no specific antidote for overdose with piracetam. Treatment for an overdose will be symptomatic treatment and may include hemodialysis. The extraction efficiency of the dialyzer is 50 to 60% for piracetam.

Storage: Store in a dry place below 30°C.

Shelf-life: Three years from the date of manufacture

Packing

Ceretam 800mg Film-Coated Tablet : PVC/Aluminium blisterstrip pack, 9 x 10's.
Ceretam Oral Solution 20%w/v : Amber PET 120ml

Manufacturer & Product registration holder

AV Manufacturing Sdn. Bhd. (667760-V)
Lot 10621 (PT 16700), Jalan Permata 2, Arab Malaysian Industrial Park,
71800 Nilai, Negeri Sembilan, Malaysia.

KEEP OUT OF REACH OF CHILDREN.

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